Egocentric Open-Vocabulary Video Question Answering with a Hybrid Knowledge-Enhanced Answer Graph

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Egocentric video question answering (EgoVQA) aims to enable intelligent agents to understand users' dynamic first-person environments, and has attracted growing research interest alongside the proliferation of wearable devices. Unlike third-person recordings, egocentric footage captures open-ended activities that vary considerably across individuals and environments, giving rise to a diverse answer space; existing graph-based open-vocabulary methods, however, rely on a single distributional signal—typically word-embedding similarity—and fail to capture the heterogeneous object–action associations characteristic of egocentric domains. We construct EGTEA-OV, an egocentric open-vocabulary benchmark built upon EGTEA Gaze+ comprising approximately 47K question–answer pairs verified through semantic filtering and manual review, with the answer vocabulary partitioned into Base, Common, Rare, and Unseen categories for fine-grained generalization evaluation. We propose EOVQA, which constructs a path-cost-guided hybrid knowledge-enhanced answer graph integrating ConceptNet, Visual Genome, GloVe, and Sentence-BERT, and applies edge-weight-guided graph attention to propagate relational context toward rare and unseen answer nodes. We evaluate EOVQA on EGTEA-OV and show that existing methods largely fail on Rare and Unseen answers, while EOVQA achieves absolute gains of 57.66 and 17.31 percentage points over the strongest baseline, with a Mean Accuracy of 82.07% across all four categories; ablation results further confirm the complementary contribution of each knowledge source.

CCS Concepts: • **Computing methodologies** → **Scene understanding**; **Information extraction**; *Knowledge representation and reasoning*; Transfer learning; • **Information systems** → *Video search.*

Additional Key Words and Phrases: egocentric video, open-vocabulary question answering, knowledge graph, graph neural network, video question answering, hand-object interaction

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1 Introduction

Egocentric video question answering (EgoVQA) asks a system to respond to natural-language queries about activities captured from a first-person viewpoint—a capability of direct relevance to wearable assistance, augmented reality, and activity-aware computing [1, 8, 28]. Unlike curated third-person recordings, egocentric footage captures continuous, unscripted daily activities from a wearer's perspective [3, 35]; the same activity unfolds differently across individuals and environments, shaped by personal habits and local contexts, giving rise to a substantially wider answer space in which the vocabulary of relevant objects, tools, and actions shifts considerably across participants and settings. This variability makes exhaustive manual annotation of all plausible answer concepts impractical and motivates the study of open-vocabulary EgoVQA, in which models must generalise to answer concepts not observed during training [9, 16].

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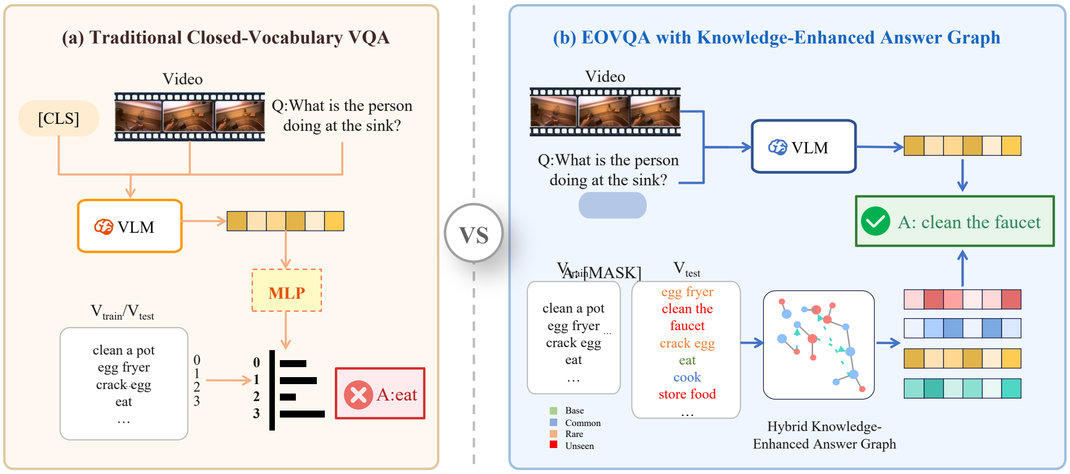

Fig. 1. Comparison between (a) traditional VLM-based approaches which map visual inputs directly to answers, often failing on unseen concepts, and (b) our proposed EOVQA framework which incorporates a Hybrid Knowledge-Enhanced Answer Graph to explicitly model relationships between seen and unseen answers.

Existing egocentric QA benchmarks address this challenge by confining evaluation to a fixed candidate set, a design choice that simplifies evaluation at the cost of open-vocabulary coverage [8, 13, 28]. EgoSchema [28] employs a five-choice multiple-choice format; QaEgo4D [8] retains fixed candidates in its multiple-choice track; and EgoCVR [13] frames the task as instance retrieval rather than free-form answer generation. What remains absent across existing egocentric QA datasets is a frequency-stratified evaluation protocol that simultaneously distinguishes among high-frequency, rare, and unseen answer categories: without such stratification, aggregate accuracy can conceal near-zero performance on rare or unseen concepts—precisely the generalisation failure that matters most in open-vocabulary settings [16, 43].

Efforts to address EgoVQA have progressed through three broad paradigms: feature-level classification, large-scale video–language pretraining, and generative instruction-tuned modelling. Early classification-based methods jointly encode per-frame visual and question features for answer prediction, achieving competitive results on closed-set benchmarks such as TGIF-QA [14] and TVQA [20], but remaining constrained by predefined label spaces. Video–language pretraining approaches, including EgoVLP [26] and All-in-One [42], learn transferable representations through contrastive or masked pretraining, yet cast QA as classification or ranking over a fixed answer vocabulary and therefore cannot handle concepts absent from the training set. Generative models such as Video-ChatGPT [27], LLaViLa [52], and Video-LLaVA [25] extend open-ended generation to egocentric video, yet systematic evaluation reveals consistent shortfalls in spatial awareness, object-interaction understanding, and coverage of domain-specific vocabulary under first-person viewpoints [1]; their outputs remain anchored to pretraining activity distributions, limiting effective coverage of rare and domain-specific answer concepts [46, 52]. Evaluating such open-form outputs against a structured answer vocabulary further requires a nearest-neighbour projection step that introduces additional sensitivity to paraphrase and surface-form variation [5, 31]. Neither closed-set classification nor unconstrained generation therefore adequately supports open-vocabulary generalisation in egocentric settings.

Structured answer-graph methods offer a complementary direction: by constructing an explicit relational graph over the answer vocabulary and applying GNN message passing to propagate information from seen to unseen answer nodes, they avoid the open-form generation problem entirely [16, 43]. Ko et al. [16] demonstrate that a GNN-based soft verbalizer—augmenting each answer node with GloVe word-vector nearest neighbours and applying graph attention to smooth answer embeddings—substantially improves rare and unseen answer prediction in third-person video QA. In egocentric settings, however, single-source distributional similarity is insufficient to capture the heterogeneous object–action associations that characterise this domain: the same answer concept simultaneously participates in functional, visual, and taxonomic relations, none of which any single distributional source covers comprehensively. To illustrate: the concept *grater* relates to *grating* via functional usage (`UsedFor`), to *cheese* via visual co-occurrence, and to *utensil* via categorical membership (`IsA`)—three distinct relation types that word-embedding similarity alone cannot jointly represent. Relying on a single distributional source therefore produces sparse graph connectivity for the compound nominal phrases and domain-specific action-object terminology characteristic of first-person activity vocabularies, leaving rare and unseen answer nodes poorly reachable. How to construct an answer graph that integrates complementary heterogeneous knowledge sources to achieve sufficient connectivity for rare and unseen egocentric concepts therefore remains an open question [16, 43].

To address this gap, this paper presents three contributions spanning dataset construction, graph construction methodology, and model design. First, we construct EGTEA-OV, an egocentric open-vocabulary VQA benchmark built upon EGTEA Gaze+ [24]—a dataset of kitchen activities recorded from 32 participants—comprising approximately 47K question–answer pairs generated by Flan-T5 [2] from action and scene annotations; quality is ensured through a Sentence-BERT-based semantic consistency filtering pipeline combined with stratified manual review by 20 trained annotators, with inter-annotator agreement verified by Cohen's κ; the answer vocabulary is partitioned into Base, Common, Rare, and Unseen categories to enable fine-grained evaluation across frequency strata. Second, and as the primary methodological contribution, we propose a *path-cost-guided hybrid graph construction algorithm* that integrates ConceptNet structured commonsense [38], Visual Genome visual co-occurrence statistics [17], GloVe lexical neighbours [33], and Sentence-BERT semantic bridges [37] into a unified answer graph; the algorithm governs multi-hop expansion through a path-cost filter that bounds traversal by accumulated edge confidence and relation-type weight, producing heterogeneous, multi-relational connectivity for rare and unseen answer nodes that no single knowledge source achieves in isolation, and constitutes a reusable module that operates independently of the downstream GNN architecture. Third, building on this graph, we propose EOVQA, which applies a two-layer graph attention network with jointly learned attention coefficients to propagate relational context toward rare and unseen answer nodes, and fuses graph-smoothed answer embeddings with a cross-modal video-question representation for answer scoring; EOVQA operate under identical backbone configurations—DeBERTa-v2-xlarge [11] for language encoding and CLIP ViT-L/14 [36] for visual encoding. Experiments on EGTEA-OV show that EOVQA achieves 90.85% Total and 82.07% Mean accuracy, with 87.77% on Rare and 31.03% on Unseen answers, representing absolute gains of 57.66 and 17.31 percentage points over the strongest adapted baseline (Ovqa+); ablation studies confirm the complementary contribution of each knowledge source, and extending the proposed graph construction approach to larger-scale egocentric benchmarks such as EPIC-KITCHENS-100 [3] and Ego4D [8] constitutes a direct avenue for future work.

2 Related Work

We organize related work along four axes that jointly inform the design of EGTEA-OV and EOVQA: video question answering, egocentric video understanding, open-vocabulary learning, and knowledge-augmented visual question answering.

2.1 Video Question Answering

Early VideoQA methods follow an encoder–decoder paradigm inherited from image VQA, where frame-level features are extracted by convolutional networks, temporal information and questions are modeled by recurrent architectures, and answers are predicted via feature fusion.

Representative benchmarks such as TGIF-QA establish spatio-temporal reasoning tasks on short video clips, covering repetition counting, action repetition, and state transition [14]. These approaches, however, struggle with long-range context and cross-event reasoning, motivating question-guided attention and memory mechanisms to highlight relevant frames or regions [20, 21].

To better capture structured interactions, graph-based methods such as Bridge to Answer and Video Graph Transformer explicitly model visual entities, relations, and temporal dynamics through message passing [32, 44]. While improving relational reasoning, these methods are primarily designed for closed-vocabulary settings and do not address generalization beyond predefined answer sets.

With the emergence of large-scale cross-modal pretraining, joint video–language representation learning has become a major paradigm for VideoQA. VideoBERT and HERO learn cross-modal representations from large-scale video–text corpora [23, 39], while ClipBERT enables efficient end-to-end learning via sparse frame sampling [19], and MERLOT Reserve further improves temporal commonsense reasoning through multimodal masked pretraining [50, 51].

More recently, instruction-tuned multimodal models have further advanced VideoQA. Video-ChatGPT adapts visual instruction tuning to temporal video understanding [27], Video-LLaVA unifies visual representations from images and videos before projection into a shared language space [25], and LaViLa leverages large-scale language supervision for egocentric video representation learning [52].

Despite these advances, recent studies show that such models still face challenges in egocentric reasoning, including spatial understanding, hand–object interaction, and intent inference, and exhibit limited coverage for rare or domain-specific concepts [1]. Across classification-based, pretraining-based, and generative approaches, a common limitation remains: existing methods lack an explicit mechanism for connecting seen answer concepts to related but unobserved ones at inference time, a gap that structured relational modeling is well positioned to address.

2.2 Egocentric Video Understanding

Egocentric videos capture continuous daily activities from a first-person perspective, introducing distinct challenges for visual representation and semantic generalization compared with third-person settings.

Large-scale datasets underpin this line of research. EPIC-KITCHENS-100 provides densely annotated kitchen activities [3], Ego4D introduces diverse tasks such as episodic memory and grounded question answering [8], and EGTEA Gaze+ offers fine-grained annotations including gaze tracking, hand masks, and action labels [24].

These data highlight key challenges: ego-motion and hand-induced occlusion complicate visual perception, while the open-ended nature of daily activities leads to substantial variability in objects and tools across environments [34, 35].

Recent benchmarks such as EgoSchema extend evaluation to long-form reasoning [28], and grounded QA tasks further require temporal evidence localization [4, 8].

However, many existing benchmarks still adopt closed answer sets or multiple-choice formats, leaving open-vocabulary answer generalization largely underexplored in egocentric VideoQA [16]. Although retrieval-augmented approaches improve long-term context modeling, they remain constrained by training-time vocabularies, limiting coverage of rare or unseen concepts [26, 40, 46].

These limitations highlight the need for benchmarks that evaluate across answer frequency strata and methods capable of reasoning beyond training-time co-occurrence patterns.

2.3 Open-Vocabulary Learning

Open-vocabulary learning aims to enable recognition or generation for categories not observed during training. CLIP exemplifies this paradigm by aligning visual and textual representations in a shared embedding space, enabling zero-shot transfer to novel categories [36].

This idea has been extended to open-vocabulary detection and segmentation through methods such as Detic and OpenSeg [7, 9, 53]. In contrast to detection, open-vocabulary visual question answering requires generating answers over an effectively unbounded concept space, rendering standard closed-set evaluation insufficient.

Addressing this challenge has taken two broad directions: leveraging retrieved or generated knowledge to expand distributional coverage, and constructing explicit structural representations of answer concept relationships. Few-shot and retrieval-augmented approaches can partially mitigate this challenge by supplying external knowledge [47], while BLIP-2 strengthens cross-modal alignment via a lightweight querying transformer bridging frozen encoders and large language models [22].

Nevertheless, these approaches do not explicitly model structured relations between seen and unseen concepts, which becomes critical when answer distributions are sparse or domain-specific [29, 30]. Structured knowledge augmentation offers a complementary direction by introducing explicit relational connections among answer concepts.

2.4 Knowledge-Augmented Visual Question Answering

Knowledge-augmented VQA has been extensively studied in the image domain, demonstrating that external knowledge improves answer coverage beyond visual evidence alone. OK-VQA highlights the need for world knowledge beyond perceptual input [30], while KAT and KRISP integrate structured knowledge graphs—such as ConceptNet [38]—with neural representations to support compositional reasoning [10, 29].

In video settings, methods such as HGA and HCRN employ graph-based reasoning to model interactions among visual entities, temporal events, and question semantics [15, 18]. However, these approaches construct graphs over scene entities rather than answer concepts and do not address generalization beyond the training vocabulary.

More closely related to our work, Ko et al. [16] propose a GNN-based soft verbalizer for open-vocabulary VideoQA, where an answer graph is constructed by augmenting each answer node with nearest-neighbor words retrieved via GloVe embeddings [33], iteratively expanding to two-hop neighborhoods; a modified graph attention network then smooths answer embeddings through neighborhood aggregation, improving performance on rare and unseen answers.

While effective, this approach relies solely on distributional statistics and does not incorporate structured commonsense knowledge (e.g., ConceptNet) or visual co-occurrence patterns characteristic of egocentric activities. Wang et al. [43] take a different approach, proposing VQA-GNN for image-level visual commonsense reasoning by combining a scene graph with a ConceptNet concept graph and performing bidirectional GNN fusion; however, this work targets

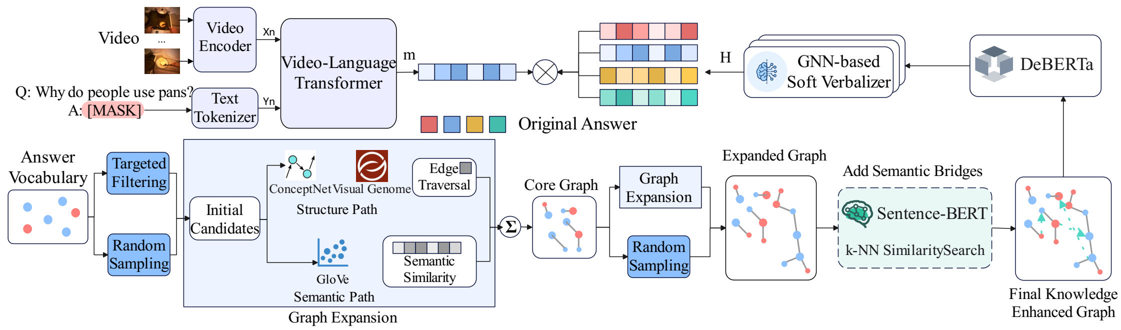

Fig. 2. Overview of the proposed EOVQA framework. The video encoder (CLIP ViT-L/14 [36]) extracts frame-level visual features, which are linearly projected and prepended as visual prefix tokens to the Video-Language Transformer for cross-modal encoding; the masked-token vector m is extracted from the Video-Language Transformer output. Simultaneously, the model constructs a Hybrid Knowledge-Enhanced Answer Graph by integrating ConceptNet, Visual Genome, GloVe-based lexical neighbors, and Sentence-BERT semantic bridges. The final reasoning is performed via a GNN-based Soft Verbalizer that fuses graph-enhanced answer embeddings H with the cross-modal representation m to predict open-vocabulary answers.

image VQA on fixed answer sets and builds its graph over scene entities rather than answer candidates, and does not address generalization to unseen answer concepts.

Taken together, these limitations motivate explicit modeling of answer relations, integration of heterogeneous knowledge sources, and evaluation on egocentric open-vocabulary benchmarks such as EGTEA-OV.

3 Method

This section presents EGTEA-OV, an egocentric open-vocabulary VQA dataset, and EOVQA, a method specifically designed for egocentric open-vocabulary video question answering. The core of EOVQA is a path-cost-guided hybrid knowledge-enhanced answer graph that integrates structured commonsense, visual co-occurrence, and multilevel semantic similarity as complementary relational sources, enabling improved prediction of rare and unseen answers. Section 3.1 describes the construction of EGTEA-OV; Section 3.2 presents the hybrid knowledge-enhanced answer graph; Section 3.3 details the EOVQA architecture and GNN-based answer prediction.

3.1 EGTEA-OV Dataset Construction

EGTEA-OV is constructed from EGTEA Gaze+, a dataset collected from 32 participants performing 86 cooking tasks in kitchen environments, comprising 28 hours of video at 1280×960 resolution and 24 Hz frame rate, with per-frame gaze information and 10,325 fine-grained action instances spanning 106 classes in a verb–noun format [24]. Its dense annotation of hand-object interactions in complex dynamic scenes makes it a suitable foundation for egocentric open-vocabulary VQA.

To extract video features, we uniformly sample T frames from each video clip; for the t-th sampled frame ($t = 1, \ldots, T$), we extract a feature vector $v_t \in \mathbb{R}^{D_v}$ using a pretrained CLIP ViT-L/14 model [36], where D_v denotes the visual feature dimension. All frame features are stacked in temporal order to form the video feature matrix $\mathbf{V} = [v_1, v_2, \ldots, v_T]^\top \in \mathbb{R}^{T \times D_v}$, with each frame vector L_2-normalised prior to construction:

$$\tilde{v}_t = \frac{v_t}{\|v_t\|_2}, \tag{1}$$

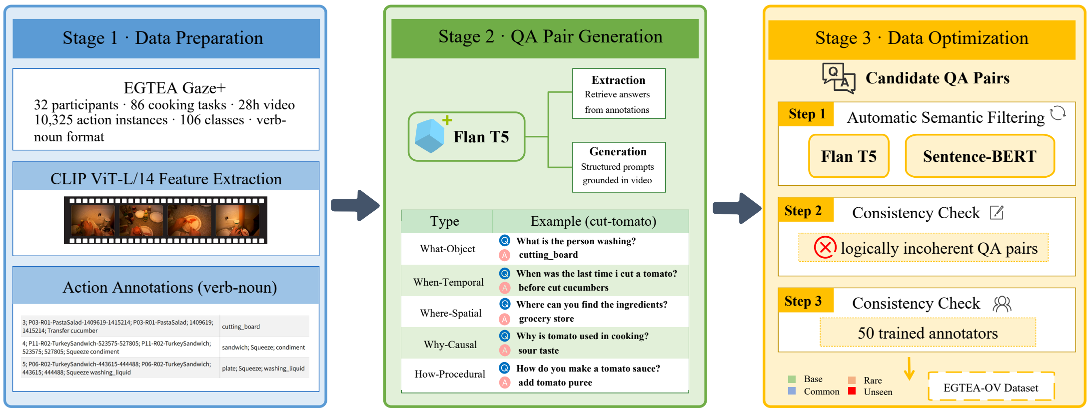

Fig. 3. The construction pipeline of the EGTEA-OV dataset. It consists of three main stages: Data Preparation (video sampling and annotation processing), QA Pair Generation (using Flan-T5 for prompt-based generation), and Data Optimization (semantic filtering and consistency checks) to ensure high-quality open-vocabulary QA pairs.

ensuring consistent scale across frames [19, 36].

For text processing, raw action annotations are first normalised to ensure input consistency; each annotation follows a verb–noun format such as `cut-tomato`, providing clear semantic structure for downstream QA generation. As shown in Fig. 3, QA pair generation follows two parallel paths driven by Flan-T5 [2]: answer extraction, which retrieves candidate answers directly from action annotations, and answer generation, which uses structured prompts to produce new answers grounded in the video content. This pipeline covers five question types—What-Object, When-Temporal, Where-Spatial, Why-Causal, and How-Procedural—so that the dataset captures a broad range of interaction semantics. Automatic generation is complemented by targeted manual checks to preserve real visual grounding [20, 30].

Candidate QA pairs undergo a three-step data optimisation pipeline before entering the final dataset, as illustrated in Fig. 3. First, automatic semantic filtering applies a generate-then-filter strategy: Flan-T5 generates a textual description d of each clip's visual content as the semantic reference [2], and Sentence-BERT encodes both d and the candidate answer a into dense vectors $e_d, s_a \in \mathbb{R}^{D_s}$ [37], where D_s denotes the sentence embedding dimension; semantic consistency is then measured by their cosine similarity:

$$s_{\text{match}}(d, a) = \text{sim}(e_d, s_a), \tag{2}$$

and pairs with $s_{\text{match}} < \tau$ are flagged as inconsistent; we set $\tau = 0.35$, selected on a held-out development subset. Second, a consistency check further removes QA pairs where the question and answer are logically incoherent or where the answer cannot be inferred from the video content. Third, flagged pairs from both prior stages are submitted to manual review by 20 trained annotators; pairs judged inconsistent or semantically redundant are removed [20, 30]. Inter-annotator agreement is assessed by Cohen's κ ($\kappa = 0.76$, indicating substantial agreement), confirming the reliability of the manual verification stage. A notable risk in LLM-based QA generation is factual hallucination [5, 31]; the generate-then-filter strategy addresses this directly: the semantic consistency check removes candidate pairs whose answers are incoherent with the video description, while the manual review stage provides a safeguard against residual errors that automated scoring cannot reliably detect, treating Flan-T5 outputs as candidates subject to independent verification rather than as ground truth.

Using the filtered data, all answers are collected into a vocabulary A with training frequency $f(a)$ for each $a \in A$; letting A_{test} denote the answers appearing in the test set, the vocabulary is partitioned into four categories according to frequency boundaries empirically determined from the training set distribution:

$$\text{Category}(a) = \begin{cases} \text{Base}, & f(a) > 600, \\ \text{Common}, & 200 \leq f(a) \leq 600, \\ \text{Rare}, & 1 \leq f(a) < 200, \\ \text{Unseen}, & f(a) = 0 \text{ and } a \in A_{\text{test}}. \end{cases} \tag{3}$$

This partition enables fine-grained assessment of model generalisation across answer frequency strata, covering high-frequency, medium-frequency, low-frequency, and unseen answers under open-vocabulary conditions. The resulting EGTEA-OV dataset spans five question types and four answer frequency categories, providing a controlled protocol for evaluating model generalisation to rare and unseen answers.

3.2 Hybrid Knowledge-Enhanced Answer Graph

Prior work on graph-based open-vocabulary VideoQA typically relies on GloVe nearest neighbours to augment answer representations [16]. In egocentric settings, however, the associations between actions, objects, and tools are highly context-dependent and shaped by individual habits and local environments, making distributional word statistics alone insufficient to bridge seen and unseen answer concepts [34, 35].

For a rare or unseen answer node to be predicted correctly, at least one path from question keywords to that node must exist in the constructed graph within the allowed hop limit $K_{\max}$. A single relational source covers only one facet of inter-concept connectivity—distributional co-occurrence, functional structure, or visual grounding—and concepts whose relevant relations are underrepresented by that source will have sparse or zero connectivity under it alone. When the constituent edge sets are coverage-complementary, their union provides non-decreasing Answer Reachability:

$$\text{AR}(E_1 \cup \cdots \cup E_m) \;\geq\; \max_i \text{AR}(E_i). \tag{4}$$

The degree to which this bound is strict depends on the extent of path-coverage overlap between sources; we analyse this empirically in the ablation study of Section **??**.

To address the single-source limitation, we construct a hybrid knowledge-enhanced answer graph by integrating four complementary sources: structured commonsense from ConceptNet [38], visual co-occurrence statistics from Visual Genome [17], word-level semantic neighbours from GloVe [33], and sentence-level semantic similarity from Sentence-BERT [37]. These sources are combined through a three-phase construction pipeline that progressively expands answer graph coverage, with targeted enhancement for concepts in the Rare and Unseen categories.

Graph construction begins with seed node selection from the answer vocabulary $\mathcal{A}$. Seeds are drawn via two parallel strategies: targeted filtering selects answer strings from the Rare and Unseen categories as defined by training frequency in Section 3.1—only the answer strings themselves are used, not any ground-truth label assignments—to prioritise concepts most likely to benefit from relational enrichment; random sampling draws additional terms from $\mathcal{A}$ to ensure broader coverage beyond high-frequency answers. The combined seed set $\mathcal{S}_0 = \mathcal{T} \cup \textsc{Rand}(\mathcal{A}, N_0)$, where $\mathcal{T}$ denotes the set of targeted Rare/Unseen answer strings, forms the starting point for graph expansion.

Starting from $\mathcal{S}_0$, candidate neighbours are retrieved from knowledge sources and accepted according to a path cost criterion. For each relation edge e in the traversal, a base weight $r(e)$ is assigned according to the relation type (Table 1), and the path cost of a multi-hop path P is defined as:

$$C(P) = \sum_{e \in P} \frac{1}{r(e)}, \quad (5)$$

where higher-confidence relation types yield lower traversal cost; a candidate node is accepted only if the accumulated path cost remains below a maximum threshold $C_{\max}$ and the total hop count does not exceed $K_{\max}$, while per-node neighbour limits and a global maximum graph size prevent uncontrolled growth. Two expansion paths operate in parallel: the Structure Path traverses ConceptNet assertions (e.g., `IsA`, `PartOf`) and Visual Genome object-relation annotations, with ConceptNet triples filtered by a minimum confidence threshold and frequency-based pruning [38], and Visual Genome co-occurrences converted to relation edges with associated confidence scores that provide visually grounded evidence [17]; the Semantic Path retrieves GloVe word-vector neighbours for all candidate nodes [33], loaded prior to structural expansion to provide early-stage semantic coverage for concepts that may lack structural connections. The outputs of both paths are merged by taking the union of node and edge sets, with duplicate edges resolved by retaining the highest-confidence instance, yielding the Core Graph. A subsequent graph expansion stage—combining edge-guided neighbour traversal with random sampling across unexpanded vocabulary terms—produces the Expanded Graph, further increasing node reachability across the answer vocabulary.

Table 1. Relation types and their base weights $r(e)$ used in graph construction. Values are determined by grid search on the development set and serve as initialisations for the learnable relation-type bias parameters $\phi_t^{(l)}$ introduced in Section 3.3; higher weights reflect stronger prior relational confidence and yield lower path traversal cost.

Source	Relation Type	Base Weight $r(e)$
ConceptNet	Synonym, IsA	1.00
	PartOf, SimilarTo	0.90
	UsedFor, HasProperty	0.80
	AtLocation	0.70
	RelatedTo	0.60
Visual Genome	Co-occurrence, Attributes	0.80–0.90
GloVe	Word-vector neighbour	0.95
Sentence-BERT	Semantic bridge (Phase 3)	$1 - d_{ij}$

The Expanded Graph is further refined by adding targeted Sentence-BERT semantic bridges [37]: for each concept in $\mathcal{T}$ that has been incorporated into the graph, its k nearest neighbours are retrieved by cosine similarity in the Sentence-BERT embedding space, and high-confidence semantic edges are added bidirectionally. This phase ensures that Rare and Unseen answer concepts are connected to semantically proximate nodes even when structural knowledge sources provide no direct path.

Per-edge weights are computed as:

$$w_{ij} = c_{ij} \cdot r_{t(e)}, \quad (6)$$

where c_{ij} is the source confidence (for Sentence-BERT edges, $c_{ij} = 1 - d_{ij}$, with d_{ij} the cosine distance), and $r_{t(e)}$ is the relation base weight from Table 1; this multiplicative form ensures that both evidence quality and relation type jointly determine edge strength. These edge weights guide the path-cost filter and candidate ranking during graph construction; in the downstream GNN soft verbalizer, attention is additionally modulated by learnable relation-type bias parameters initialised from these weights, as described in Section 3.3. After assembling the weighted adjacency matrix $\tilde{A}$ augmented with self-loops, symmetric normalisation is applied:

$$\hat{A} = \tilde{D}^{-1/2}\,\tilde{A}\,\tilde{D}^{-1/2}, \tag{7}$$

where $\tilde{D}$ is the degree matrix of $\tilde{A}$.

Each graph node is represented by the first-token ([CLS]) output of DeBERTa-v2-xlarge [11]; GloVe embeddings are used as a fallback when contextualised DeBERTa encoding is unavailable. The resulting initial feature matrix $X^{(0)} \in \mathbb{R}^{N\times D}$ (with $D = 1536$) is the input to the GNN soft verbalizer.

3.3 EOVQA Answer Prediction

The answer prediction module of EOVQA comprises three components: cross-modal video-question encoding, GNN-based answer embedding refinement via the hybrid knowledge-enhanced answer graph, and similarity-based answer scoring with cross-entropy training. The language encoder is DeBERTa-v2-xlarge [11] and the visual encoder is CLIP ViT-L/14 [36].

The video encoder extracts frame-level visual features $\{v_t\}_{t=1}^{T} \in \mathbb{R}^{T\times D_v}$; a learnable linear projection maps each frame embedding into the DeBERTa hidden space $\mathbb{R}^{D}$, and the resulting visual tokens are prepended as a prefix to the tokenised question sequence $Y_n = \{q_u\}_{u=1}^{U}$. Both are forwarded through the Video-Language Transformer to produce a fused sequence representation:

$$Z = \mathrm{CM}(\mathbf{V}, Q) \in \mathbb{R}^{(T+U)\times D}, \tag{8}$$

where $D = 1536$ denotes the hidden dimension; the masked-token vector $m = Z_{[\mathrm{MASK}]} \in \mathbb{R}^{D}$ carries joint context from the video and question and serves as the query for answer matching. We freeze most pretrained language-model parameters and train adapter modules [12] (bottleneck downsampling factor 8) together with the visual projection layer to adapt the cross-modal encoder to egocentric video-question inputs.

Answer node embeddings are extracted from the first-token ([CLS]) representation of DeBERTa-v2-xlarge (hidden size $D = 1536$), with the answer encoder frozen throughout training [16]. To align graph node features with the cross-modal hidden space and enable graph-space feature adaptation, a learnable linear projection $\mathbf{W}_{\mathrm{proj}} \in \mathbb{R}^{D\times D}$ is applied before GNN input:

$$x_i^{(0)} = \mathbf{W}_{\mathrm{proj}}\, v_i^{\mathrm{DeBERTa}} \in \mathbb{R}^{D}, \tag{9}$$

yielding the initial node feature matrix $V_{\mathrm{ans}} = \{x_i^{(0)}\}_{i=1}^{N} \in \mathbb{R}^{N\times D}$ that serves as input to the GNN soft verbalizer.

We adopt a two-layer ($L = 2$) modified graph attention network [16, 41] that smooths answer embeddings by aggregating information from neighbouring answer concepts. To allow the model to adapt the structural priors encoded in Table 1 to the target domain, we introduce a learnable relation-type bias $\phi_{t(e_{ij})}^{(l)} \in \mathbb{R}$ for each relation type $t \in \mathcal{T}$ and layer $l \in \{1, 2\}$, initialised from the corresponding base weight $r_{t(e)}$ in Table 1. This bias is added to the dot-product attention score before the nonlinearity, enabling the GNN to adjust the relative importance of different relational sources during training while retaining the knowledge-source prior as initialisation. The attention coefficient from node j to node i at layer l is computed as:

$$\alpha_{ij}^{(l)} = \frac{\exp\Big(\mathrm{LeakyReLU}_{0.2}\Big(\big(\mathbf{W}_{\mathrm{dst}}^{(l)}\, h_i^{(l-1)}\big)^{\top}\big(\mathbf{W}_{\mathrm{src}}^{(l)}\, h_j^{(l-1)}\big) + \phi_{t(e_{ij})}^{(l)}\Big)\Big)}{\sum_{k\in\mathcal{N}(i)} \exp\Big(\mathrm{LeakyReLU}_{0.2}\Big(\big(\mathbf{W}_{\mathrm{dst}}^{(l)}\, h_i^{(l-1)}\big)^{\top}\big(\mathbf{W}_{\mathrm{src}}^{(l)}\, h_k^{(l-1)}\big) + \phi_{t(e_{ik})}^{(l)}\Big)\Big)}, \tag{10}$$

where $\mathbf{W}_{\text{src}}^{(l)}, \mathbf{W}_{\text{dst}}^{(l)} \in \mathbb{R}^{D\times D}$ are learnable source- and destination-side projection matrices. The node update rule is:

$$h_i^{(l)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(l)}\, \mathbf{W}_{\text{src}}^{(l)}\, h_j^{(l-1)}\right), \tag{11}$$

where each node weights neighbour contributions according to learned pairwise compatibility, propagating relational information across the answer graph [16, 41]. After $L = 2$ layers of message passing, updated node embeddings $H^{(L)} \in \mathbb{R}^{N\times D}$ are obtained, capturing information up to two-hop neighbourhoods in the answer graph.

To balance the original pretrained semantics of answer embeddings with graph-propagated relational information, a convex interpolation is applied [16]:

$$\widehat{H} = \varepsilon\, V_{\text{ans}} + (1-\varepsilon)\, H^{(L)}, \tag{12}$$

where $\varepsilon \in [0,1]$ is a fixed interpolation coefficient selected on the validation set. The answer encoder is frozen during training so that the GNN soft verbalizer does not alter the original word semantics of the initial embeddings [16].

At inference, the masked-token vector m is matched against all answer embeddings via dot product $s_i = m^\top \widehat{h}_i$, and the softmax distribution over the answer vocabulary is:

$$p_i = \frac{\exp(s_i)}{\sum_{k=1}^{N} \exp(s_k)}. \tag{13}$$

The training objective is the cross-entropy loss over the ground-truth answer index a_{GT}:

$$\mathcal{L} = -\log p_{a_{\text{GT}}} = -\log \frac{\exp(m^\top \widehat{h}_{a_{\text{GT}}})}{\sum_{k=1}^{N} \exp(m^\top \widehat{h}_k)}. \tag{14}$$

By jointly optimising the visual projection layer, the adapter modules, and the GNN parameters ($\mathbf{W}_{\text{proj}}$, $\mathbf{W}_{\text{src}}^{(l)}$, $\mathbf{W}_{\text{dst}}^{(l)}$, $\{\phi_t^{(l)}\}_{t\in\mathcal{T}, l\in\{1,2\}}$) while keeping the pretrained language model backbone and the answer encoder frozen, EOVQA jointly leverages video-question semantics and structured answer-graph knowledge to predict open-vocabulary answers [16, 46].

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets and Baselines. We evaluate EOVQA on the EGTEA-OV dataset, which contains QA triplets spanning five question types with answers partitioned by training frequency into Base, Common, Rare, and Unseen categories [24]. We compare against six methods. **OVQA** [16] is the GNN-based open-vocabulary VideoQA framework of Ko et al., which builds a GNN soft verbalizer upon the FrozenBiLM masked text infilling backbone [46]; we reproduce the GloVe-based answer graph construction following the methodology and source code provided by the authors and apply the framework to EGTEA-OV. **Ovqa+** [16] denotes the domain-adapted variant of OVQA evaluated under the same EGTEA-OV protocol. **Video-ChatGPT** [27] adapts large vision-language models to video understanding via visual instruction tuning. **SeViLA** [48] applies self-chained visual-language alignment for temporal video grounding and question answering. **LLaViLa** [52] integrates large language models with egocentric visual representations through large-scale narration supervision. **Video-LLaVA** [25] extends LLaVA to video understanding by aligning visual representations before language model projection.

EGTEA-OV comprises 38,499 training and 9,347 test QA pairs across five question types; Table 2 summarizes the answer vocabulary statistics. *How* questions account for 45.9% of training instances and *Why* questions for 30.7%,

Table 2. Answer vocabulary statistics for EGTEA-OV.

Category	Criterion	# Unique Answers
Base	$f(a) > 600$	8
Common	$200 \leq f(a) \leq 600$	31
Rare	$1 \leq f(a) < 200$	289
Unseen	$f(a) = 0$	29
Total	–	357

together exceeding 75% of the dataset, reflecting the procedural and causal nature of kitchen activity narrations. The test split mirrors this distribution closely, with *How* and *Why* questions accounting for 46.4% and 30.5%, respectively.

Existing VideoQA benchmarks such as ActivityNet-QA [49] and MSRVTT-QA [45] operate under closed-set answer vocabularies without frequency-based partitioning, and therefore do not directly evaluate rare or unseen answer generalization under egocentric settings. As third-person video datasets, their answer spaces also differ from the first-person hand-object interaction vocabulary of egocentric cooking activities, which limits their applicability to the evaluation task addressed in this work. EGTEA-OV is constructed to fill this gap by providing a stratified open-vocabulary evaluation protocol grounded in first-person cooking scenarios.

4.1.2 Implementation Details. The framework uses `deberta-v2-xlarge` [11] as the language backbone and CLIP ViT-L/14 [36] as the video encoder, sampling up to 10 frames per clip with a maximum input length of 256 tokens. Training runs for 10 epochs using Adam (lr $= 3 \times 10^{-4}$, batch size 32) on four NVIDIA A100 GPUs with mixed precision. The GNN module uses two GAT-style layers [41] via `torch_geometric` [6]; the interpolation coefficient ε is selected from $\{0.5, 0.6, 0.7, 0.8, 0.9\}$ on the development set according to Mean Accuracy. Adapter modules [12] with bottleneck factor 8 are inserted into transformer sublayers; the pretrained backbone and answer encoder remain frozen throughout, with only the visual projection layer, adapter weights, and GNN parameters updated.

For generative baselines (Video-ChatGPT, SeViLA, LLaViLa, and Video-LLaVA), each generated output is lowercased and stripped of stopwords, then matched to the nearest candidate in the EGTEA-OV answer vocabulary by Sentence-BERT cosine similarity [37]; this nearest-neighbor mapping yields the predicted answer for exact-match scoring.

4.1.3 Evaluation Metrics. We evaluate both QA performance and answer-graph quality using complementary metrics.

QA Performance Metrics. For open-vocabulary QA we report two accuracy measures. *Total Accuracy* is the fraction of correctly answered questions over the full evaluation set S_{QA}:

$$\mathrm{Acc}_{\mathrm{total}} = \frac{1}{|S_{\mathrm{QA}}|} \sum_{q \in S_{\mathrm{QA}}} \mathbf{1}\{\hat{a}_q = a_q\} \times 100, \tag{15}$$

where a_q is the ground-truth answer and $\hat{a}_q$ is the prediction. *Mean Accuracy* averages per-class accuracy to reduce bias from frequent answers:

$$\mathrm{Acc}_{\mathrm{mean}} = \frac{1}{C} \sum_{c=1}^{C} \frac{1}{|S_c|} \sum_{q \in S_c} \mathbf{1}\{\hat{a}_q = a_q\} \times 100, \tag{16}$$

where C denotes the number of answer classes; these measures follow common practice in VideoQA and open-vocabulary evaluation [14, 16].

The nearest-neighbor canonicalization applied to generative outputs follows the evaluation convention established for open-vocabulary VideoQA, where free-form predictions are projected onto a predefined candidate set to enable exact-match scoring [16]; adopting Sentence-BERT for this projection is consistent with the semantic representation used throughout the answer graph construction pipeline [37].

Answer-Graph Quality Metrics. We treat the following measures as task-oriented structural diagnostics for the constructed answer graph, each targeting a distinct structural property relevant to open-vocabulary answer prediction.

Node Coverage (NC) is the percentage of candidate answers that appear as nodes in the graph:

$$\mathrm{NC} = \frac{|\mathcal{V}_A \cap \mathcal{V}_G|}{|\mathcal{V}_A|} \times 100, \tag{17}$$

where $\mathcal{V}_A$ is the candidate answer set and $\mathcal{V}_G$ is the graph node set. Since the construction pipeline is designed to incorporate every candidate answer string as a node, NC functions primarily as a completeness check rather than as a discriminative metric.

Semantic Consistency (SC) measures the average cosine similarity between Sentence-BERT embeddings of connected node pairs [37]:

$$\mathrm{SC} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \mathrm{sim}(e_i, e_j), \tag{18}$$

where $\mathcal{E}$ is the edge set and e_i, e_j are the Sentence-BERT embeddings of nodes i and j; higher values indicate more semantically coherent edges.

Answer Reachability (AR) is the percentage of QA pairs for which the answer node is reachable from at least one question keyword node within a fixed hop limit H:

$$\mathrm{AR} = \frac{1}{|S_{\mathrm{QA}}|} \sum_{q\in S_{\mathrm{QA}}} \mathbf{1}\Big\{\exists\, v_k \in \mathcal{K}_q,\ d_G(v_k,\, v_{a_q}) \le H\Big\} \times 100, \tag{19}$$

where $\mathcal{K}_q$ is the set of keyword nodes extracted from question q, v_{a_q} is the answer node, and $d_G(\cdot,\cdot)$ denotes the shortest-path distance in the graph. This metric captures whether the constructed graph provides a usable multi-hop route from question evidence to answer concepts. *Low-similarity Edge Rate* (LE) measures the proportion of graph edges whose Sentence-BERT cosine similarity between the embeddings of connected node pairs falls below a threshold δ:

$$\mathrm{LE} = \frac{\Big|\{(i,j) \in \mathcal{E} : \mathrm{sim}(e_i, e_j) < \delta\}\Big|}{|\mathcal{E}|} \times 100, \tag{20}$$

where $\mathcal{E}$ is the edge set, e_i, e_j are the Sentence-BERT embeddings of nodes i and j, and $\delta = 0.5$ following the evaluation pipeline [37]; lower values indicate fewer semantically weak connections in the graph. Together, these metrics provide a compact view of coverage, semantic coherence, and reasoning accessibility in the answer graph.

4.2 Comparative Evaluation

Table 3 reports performance of EOVQA and all baselines on EGTEA-OV, where B, C, R, U, T, and M denote Base, Common, Rare, Unseen, Total, and Mean accuracy, respectively.

OVQA, which implements Ko et al.'s GNN soft verbalizer atop the FrozenBiLM backbone [16, 46] and is evaluated on EGTEA-OV following the reproduced graph construction, achieves 20.21% Total and 9.90% Mean accuracy, with near-zero performance on Rare (0.24%) and Unseen (0.03%), indicating limited transferability of the GloVe-only answer graph to the egocentric cooking vocabulary in EGTEA-OV. Ovqa+, which adapts the same framework through graph-construction modifications and model-level tuning on EGTEA-OV, raises Total accuracy to 47.83% and Mean to 27.59%, suggesting that domain adaptation substantially benefits the original single-source graph setup. Nevertheless, Ovqa+ remains far below EOVQA on Rare (30.11% vs. 87.77%) and Unseen (13.72% vs. 31.03%), which is consistent with the view that lexical neighbors alone are insufficient for compound nominal phrases and fine-grained utensil terminology in EGTEA-OV [33, 35]. Within the EOVQA architecture, the GloVe-only ablation (Table 4) achieves 74.43% Total, while the full hybrid configuration reaches 90.85% Total, supporting the conclusion that the additional gains come from multi-source graph construction rather than from the backbone alone.

Among generative baselines, LLaViLa [52] records the highest Base accuracy (92.08%) but collapses to near-zero on Rare (1.35%) and Unseen (0.18%), yielding 48.01% Total. LLaViLa is pretrained on Ego4D narrations [8] and therefore produces fluent descriptions of frequent egocentric concepts; after nearest-neighbor canonicalization, its outputs align better with high-frequency answers than with rare or unseen answer forms that are underrepresented in the pretraining distribution [1]. The same pattern—strong Base performance alongside much weaker Rare and Unseen scores—appears in the other generation-based methods as well (Video-ChatGPT: 8.55% Rare, 3.45% Unseen; SeViLA: 12.43% Rare, 4.11% Unseen), indicating a consistent difficulty in matching free-form generations to the EGTEA-OV candidate vocabulary under exact-match evaluation [46].

EOVQA achieves 87.77% on Rare and 31.03% on Unseen by constructing multi-hop reasoning paths that jointly exploit visual co-occurrence, structured commonsense, and lexical similarity.

Table 3. Performance comparison on EGTEA-OV. B, C, R, U, T, and M denote Base, Common, Rare, Unseen, Total, and Mean accuracy (%), respectively. OVQA denotes the framework of Ko et al. [16] reproduced with the GloVe-based answer graph construction following the original methodology; Ovqa+ denotes the adapted variant evaluated under the same protocol on EGTEA-OV.

Model	B (%)	C (%)	R (%)	U (%)	T (%)	M (%)
OVQA [16]	30.41	6.33	0.24	0.03	20.21	9.90
Ovqa+ [16]	74.59	46.26	30.11	13.72	47.83	27.59
Video-ChatGPT [27]	43.22	29.37	8.55	3.45	27.30	21.15
SeViLA [48]	45.52	28.15	12.43	4.11	35.80	22.55
LLaViLa [52]	92.08	56.77	1.35	0.18	48.01	37.59
Video-LLaVA [25]	38.85	30.41	21.75	10.39	30.35	25.35
EOVQA (Ours)	**94.98**	**90.26**	**87.77**	**31.03**	**90.85**	**82.07**

4.3 Ablation Studies

We perform ablation studies to isolate the contributions of three key components: knowledge sources, Sentence-BERT semantic smoothing, and GNN message passing.

4.3.1 Impact of Knowledge Sources. Table 4 and Table 5 report QA performance and graph quality metrics across knowledge source configurations. All configurations share the full EOVQA architecture and differ only in the sources used to construct the answer graph.

The contribution of each source reflects its complementary coverage of the egocentric cooking domain. Visual Genome encodes visually grounded co-occurrence patterns characteristic of kitchen activities, connecting objects and tools observed in egocentric frames to their associated actions and functions; used in isolation, it achieves the strongest single-source result (83.66% Total, 73.47% Mean), reflecting its alignment with the object- and action-centric questions that dominate EGTEA-OV [17, 35]. ConceptNet supplies commonsense relations—`IsA`, `PartOf`, `UsedFor`, `AtLocation`—that connect visual observations to semantic action concepts and functional categories absent from purely distributional sources [38]. GloVe broadens lexical coverage by connecting surface-form variants and terminological synonyms across the answer vocabulary [33]. Among the dual-source configurations, all three substantially improve Rare accuracy over the best single-source baseline, while VG+CON additionally achieves the strongest Unseen result (31.03%) among dual-source configurations; neither combination, however, fully matches the three-source configuration in Answer Reachability (67.50%) or Rare accuracy (87.77%). Combining all three sources yields the best results across all categories (90.85% Total, 82.07% Mean, 87.77% Rare, 31.03% Unseen).

The graph quality metrics in Table 5 show a consistent structural pattern across configurations. Node Coverage reaches 100% in all cases, confirming that the construction pipeline incorporates every candidate answer as a graph node. Semantic Consistency and Answer Reachability exhibit a trade-off: the full three-source configuration achieves the highest AR (67.50%) while recording a lower SC (56.33) than Visual Genome alone (62.16), because heterogeneous cross-source edges broaden graph coverage but reduce average pairwise semantic similarity across all edges. In this setting, the AR gain over VG alone aligns with the improvement in Rare and Unseen accuracy, which suggests that increased reachability is associated with better propagation toward rare and unseen answer nodes. At the same time, the decrease in SC indicates that adding heterogeneous knowledge sources introduces more diverse connections, so the edge-weight-guided attention mechanism becomes important for down-weighting less informative edges during message passing [41].

Table 4. Ablation study on knowledge sources for answer graph construction on EGTEA-OV.

Configuration	B (%)	C (%)	R (%)	U (%)	T (%)	M (%)
GloVe only	81.90	82.11	58.59	24.17	74.43	44.27
VG only	83.84	86.35	81.07	27.59	83.66	73.47
CON only	82.63	82.17	59.55	24.11	74.96	45.37
GloVe + VG	92.42	89.83	86.75	27.62	89.55	81.06
GloVe + CON	90.93	89.98	86.69	27.56	89.04	80.80
VG + CON	92.66	90.16	86.65	31.03	89.70	81.66
All (Full)	**94.98**	**90.26**	**87.77**	**31.03**	**90.85**	**82.07**

Table 5. Graph quality metrics for knowledge source ablations (test set).

Configuration	AR (%)	NC (%)	SC	Low-sim Edge (%)
GloVe only	36.90	100	47.91	55.83
VG only	44.50	100	62.16	12.71
CON only	16.23	100	42.97	67.22
GloVe + VG	49.41	100	61.92	16.43
GloVe + CON	50.09	100	59.97	16.88
VG + CON	53.28	100	59.70	16.49
All (Full)	**67.50**	**100**	**56.33**	**24.52**

4.3.2 Role of Sentence-BERT Semantic Smoothing. Table 6 shows that incorporating Sentence-BERT raises Unseen accuracy from 27.56% to 31.03%. The structural changes associated with Sentence-BERT are reflected in Table 7: Semantic Consistency rises from 6.44 to 56.33, while the proportion of low-similarity edges drops from 81.97% to 24.52%. These results suggest that Sentence-BERT contributes in two ways: it reduces noisy semantic mismatches and adds targeted semantic bridges for Rare and Unseen concepts that structural sources do not directly connect.

Table 6. Ablation study on Sentence-BERT semantic smoothing.

Configuration	B (%)	C (%)	R (%)	U (%)	T (%)	M (%)
Without SBERT	88.64	88.38	86.69	27.56	88.32	80.65
With SBERT	**94.98**	**90.26**	**87.77**	**31.03**	**90.85**	**82.07**

Table 7. Graph quality metrics for Sentence-BERT ablation (test set).

Configuration	AR (%)	NC (%)	SC	Low-sim Edge (%)
Without SBERT	65.53	100	6.44	81.97
With SBERT	**67.50**	**100**	**56.33**	**24.52**

4.3.3 Analysis of GNN Message Passing. Table 8 isolates the contributions of GNN message passing through five configurations, all of which share the same hybrid knowledge-enhanced answer graph constructed with the full three-source configuration; the configurations differ exclusively in how the GNN processes and propagates node features. Removing message passing entirely (*G_noMP*) reduces Mean accuracy to 79.51%, confirming that graph-based reasoning remains important in this setting. Using mean aggregation without attention (*G_noAttn_Mean*) yields 80.14% Mean, below the full model's 82.07%, which indicates that edge-weight-guided attention helps distinguish neighbors with different relational relevance [41]. A single GNN iteration (*G_oneIter*) yields 80.42% Mean and 24.14% Unseen, suggesting that the two-layer design is more effective than a shallower propagation step for reaching answer concepts that are only indirectly connected to question-relevant nodes. Replacing initial embeddings entirely with GNN outputs (*G_replace*) degrades performance to 79.81% Mean—yet remains above G_noMP (79.51%)—which suggests that graph propagation is beneficial, while preserving the original DeBERTa embeddings still helps retain pretrained lexical semantics [16, 41]. The interpolation coefficient ε is selected from $\{0.5, 0.6, 0.7, 0.8, 0.9\}$ on the development set; across this range, GNN-smoothed embeddings consistently improve over the no-GNN baseline ($\varepsilon = 1.0$), indicating that graph-propagated context complements the original pretrained semantics [16].

4.4 Hyperparameter Sensitivity Analysis

To assess the robustness of the answer-graph construction, we conduct a hyperparameter sensitivity study on EGTEA-OV, evaluating four parameters: maximum graph size, maximum neighbors per node, minimum confidence threshold for edge inclusion, and path constraints parameterized as (MAX_PATH_COST, MAX_HOPS); results are summarized in Fig. 4, reporting both Mean Accuracy and Unseen Accuracy.

As shown in Fig. 4(a), increasing the maximum graph size from 40k to 60k yields the best performance (Mean 82.07%, Unseen 31.03%); performance declines at 80k and beyond, where Unseen accuracy drops to 27.59%, suggesting that larger graphs are associated with lower Unseen accuracy in this setting.

Table 8. Ablation study on GNN message passing mechanisms. All configurations share the same hybrid knowledge-enhanced answer graph; they differ only in the GNN message-passing mechanism applied.

Configuration	B (%)	C (%)	R (%)	U (%)	T (%)	M (%)
G_noMP	87.32	89.70	85.30	24.11	87.29	79.51
G_noAttn_Mean	91.44	89.95	86.17	24.17	89.07	80.14
G_oneIter	89.64	89.92	86.72	24.14	88.60	80.42
G_replace	88.38	89.89	85.51	24.10	87.77	79.81
G_full (Ours)	**94.98**	**90.26**	**87.77**	**31.03**	**90.85**	**82.07**

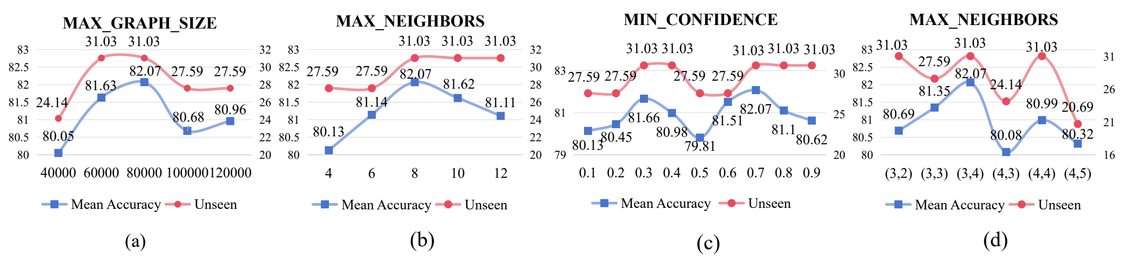

Fig. 4. Hyperparameter sensitivity analysis on the EGTEA-OV dataset. We evaluate the impact of (a) maximum graph size, (b) maximum neighbors per node, (c) minimum confidence threshold for edge inclusion, and (d) path constraints (Max_Path_Cost, Max_Hops) on Mean Accuracy and Unseen Accuracy.

For maximum neighbors per node (Fig. 4(b)), performance peaks at 8 neighbors (Mean 82.07%, Unseen 31.03%) and remains stable at 10 and 12, indicating that moderate local connectivity is sufficient for effective multi-hop reasoning in this setting.

The minimum confidence threshold (Fig. 4(c)) exhibits a non-monotonic pattern that reflects the different confidence ranges occupied by edges from different knowledge sources. In this setting, lower thresholds retain more edges but also more noisy connections, while higher thresholds filter out a larger portion of structurally useful bridging edges. The observed optimum therefore comes from balancing edge retention and edge noise rather than from a single source dominating the graph. This interaction explains why the best value is not at the extreme ends of the tested range.

For path constraints (Fig. 4(d)), parameterized as (Max_Path_Cost, Max_Hops), the configuration (3, 4) achieves the best balance (Mean 82.07%, Unseen 31.03%), while looser constraints such as (4, 5) reduce Unseen accuracy to 20.69%, indicating that bounded path search is preferable in this setting.

Across the four hyperparameters studied, performance is most sensitive to graph scale and path constraints, while the neighbor count and confidence threshold admit wider near-optimal ranges, suggesting that the construction pipeline is reasonably robust to moderate variation in these parameters.

4.5 Qualitative Analysis

Fig. 5 illustrates the reasoning process of EOVQA through a representative example, showing how three knowledge sources jointly support the prediction of an Unseen answer that the baseline model fails to identify.

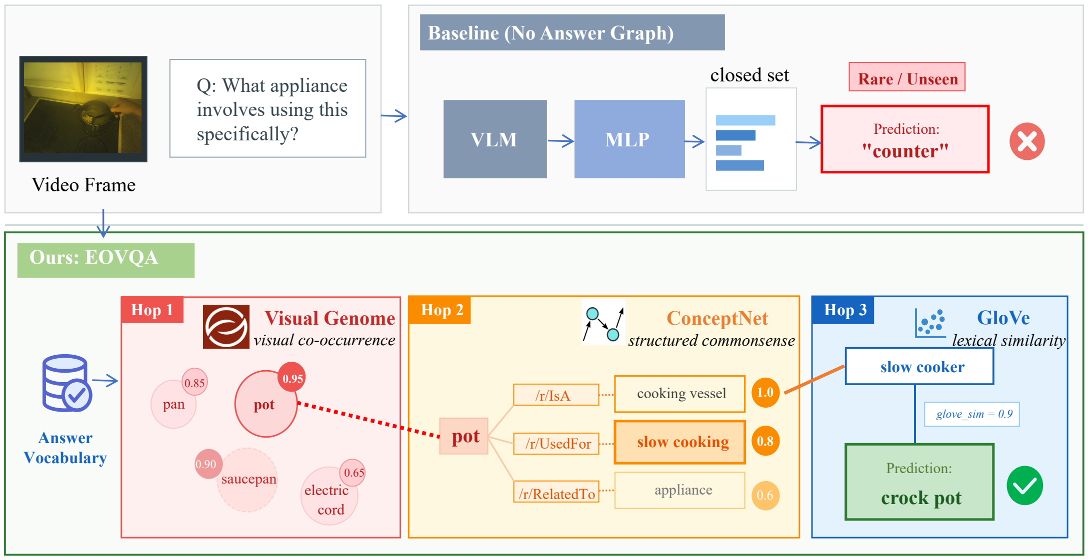

Fig. 5. Qualitative example illustrating the reasoning process of EOVQA. Given the egocentric video frame and the question "What appliance involves using this specifically?", the baseline model (VLM + MLP) without the answer graph incorrectly predicts "counter". EOVQA constructs a three-hop reasoning path through Visual Genome (visual grounding), ConceptNet (semantic relations), and GloVe (lexical similarity), correctly identifying the Unseen answer "crock pot" that does not appear in the training set.

Given the egocentric video frame and the question "What appliance involves using this specifically?", the baseline model (VLM + MLP) predicts "counter"—a visually prominent surface in the scene but semantically incorrect as an appliance answer.

EOVQA first retrieves visually grounded candidates from Visual Genome: *pot* (0.95), *saucepan* (0.90), *pan* (0.85), and *electric cord* (0.65). ConceptNet then traverses semantic relations from the highest-confidence candidate *pot*: `/r/IsA` to *cooking vessel* (1.0), `/r/UsedFor` to *slow cooking* (0.8), and `/r/RelatedTo` to *appliance* (0.6). GloVe bridges the remaining gap: *slow cooking* connects to *slow cooker* with similarity 0.9, and under GNN-smoothed scoring, *crock pot*—an Unseen answer absent from the training set—receives the highest similarity to the masked-token representation and is selected as the final prediction.

The three knowledge sources activate sequentially and non-redundantly: Visual Genome grounds the reasoning in the observed scene, ConceptNet expands into functional and categorical relations, and GloVe closes the terminological gap to the correct answer. Removing any single source weakens this chain.

A second case demonstrates the method's behavior on Rare answers within *How*-type questions, which constitute the largest question category in EGTEA-OV. Given the question "How does the person prepare the cheese for the salad?", the baseline predicts "preparing"—a high-frequency general action that does not identify the specific technique. Visual Genome detects a *cheese grater* in the frame; ConceptNet traverses `/r/UsedFor` from *cheese grater* to *grating*, a semantically specific operation absent from the high-frequency training vocabulary. EOVQA correctly predicts "grating" (Rare), whereas the baseline conflates the specific action with a generic verb due to frequency bias. This example complements the Unseen case above and shows that the method generalizes across answer frequency categories and question types through structured multi-source reasoning.

The approach nonetheless has limitations. When the visual evidence is weak and multiple answer candidates occupy similar graph neighborhoods, the model may still prefer a more frequent or more visually prominent concept. This behavior suggests that fine-grained disambiguation remains challenging in cases where the scene evidence does not clearly separate closely related answers.

5 Conclusion

Contributions. This paper addresses egocentric open-vocabulary video question answering through three contributions spanning dataset construction, method design, and empirical evaluation. We introduce EGTEA-OV, an egocentric open-vocabulary VQA benchmark built on EGTEA Gaze+ [24], comprising approximately 47K question–answer pairs across five question types. QA pair generation employs Flan-T5 [2] via two parallel paths—answer extraction and answer generation—while quality assurance combines Sentence-BERT-based semantic consistency filtering [37] with stratified manual review conducted by 50 trained annotators. The answer vocabulary is partitioned into four frequency-based categories (Base, Common, Rare, and Unseen), providing a controlled and reproducible protocol for evaluating open-vocabulary generalization. Building on this benchmark, we propose EOVQA, which constructs a hybrid knowledge-enhanced answer graph by integrating ConceptNet structured commonsense [38], Visual Genome visual co-occurrence statistics [17], GloVe lexical neighbors [33], and Sentence-BERT semantic bridges; a two-layer graph attention network with edge-weight-guided attention propagates relational context toward rare and unseen answer nodes, while an interpolation-based fusion preserves the original DeBERTa-v2-xlarge answer embeddings [11]. Experiments on EGTEA-OV show that EOVQA achieves 87.77% on Rare and 31.03% on Unseen answers, representing absolute improvements of 57.66 and 17.31 percentage points over the strongest adapted baseline (Ovqa+ [16]), and outperforms all six baselines across all four answer frequency categories.

Evaluation conclusions. Several findings of broader methodological significance emerge from the experimental analysis. First, Visual Genome consistently outperforms GloVe as a single knowledge source (83.66% vs. 74.43% Total accuracy), confirming that visually grounded co-occurrence statistics provide more relevant relational priors than distributional word statistics in hand-object interaction domains [35]. Second, combining heterogeneous knowledge sources introduces a structural trade-off between Semantic Consistency (SC) and Answer Reachability (AR): the full three-source configuration achieves the highest AR (67.50%) alongside a lower SC (56.33 vs. 62.16 for Visual Genome alone), because cross-source edges broaden topological coverage while reducing average pairwise semantic similarity; the superiority of edge-weight-guided attention over mean aggregation (+1.93% Mean) confirms that explicit relational weighting is critical when graph edge quality is heterogeneous [41]. Third, Sentence-BERT semantic bridges contribute disproportionately to Unseen performance (+3.47% absolute) with negligible impact on Base and Common categories, indicating that targeted bridging for underrepresented concepts constitutes an effective and focused intervention. Fourth, generation-based baselines—despite strong Base performance (LLaViLa: 92.08%)—collapse to near-zero on Rare (1.35%) and Unseen (0.18%) answers [52], revealing a systematic limitation of pretrained generative models in matching domain-specific egocentric vocabulary under exact-match evaluation.

Limitations. Several limitations define the scope of this work. EGTEA-OV is derived from a single dataset of controlled kitchen activities recorded by 32 participants, which may introduce domain and demographic biases that limit generalization to other egocentric settings such as outdoor activities or professional tasks. The Unseen category contains only 29 unique answer types, yielding high evaluation variance; conclusions regarding Unseen generalization should therefore be interpreted with caution, and bootstrap confidence intervals are recommended for follow-up studies. Graph construction incurs a one-time preprocessing cost proportional to the answer vocabulary size, which may become non-trivial for significantly larger answer spaces. Finally, when multiple answer candidates occupy similar graph neighborhoods, the model may still favor more frequent or visually prominent concepts, indicating that fine-grained disambiguation in ambiguous visual contexts remains an open challenge.

Future directions. Three directions follow directly from these findings. The low absolute Unseen accuracy motivates dynamic graph augmentation, in which new answer concepts encountered at inference time are incorporated as lightweight adapter nodes via few-shot embedding approximation, without requiring a full graph rebuild. The observed SC–AR trade-off suggests that jointly learning edge weights with the GNN—rather than fixing them from source confidence scores—could better balance graph coherence and topological coverage. Extending EGTEA-OV to encompass larger-scale egocentric benchmarks such as EPIC-KITCHENS-100 [3] and Ego4D [8] would provide a more demanding testbed for open-vocabulary generalization and reduce the domain-specific bias inherent in single-dataset evaluation.

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